

(Cantor) lesson 26 Cardinality of Infinite Sets
 Finite sets : $|A| = \# \text{ of elements in } A$

$|\mathbb{N}| = \infty$? $|\mathbb{Z}| = \infty$? $|\mathbb{N}| = |\mathbb{Z}|$?

Def Two sets A and B have the same cardinality (size) if there is a bijection $f: A \rightarrow B$

one-to-one correspondence

+ve integer
 ↗
 all integers

Prop $\mathbb{N}$ has the same cardinality as $\mathbb{Z}$.

$\mathbb{N}$	1	2	3	4	5	6	7	8	...
	↕	↕	↕	↕	↕	↕	↕	↕	
$\mathbb{Z}$	0	1	-1	2	-2	3	-3	4	

Proof We will show that there is a bijection

$$f: \mathbb{N} \rightarrow \mathbb{Z}, \quad f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ \frac{1-n}{2} & \text{if } n \text{ is odd} \end{cases}$$

f is 1-to-1:

If $f(n) = f(m)$, then

• both +ve : $\frac{n}{2} = \frac{m}{2} \rightarrow n = m$

• both non +ve : $\frac{1-n}{2} = \frac{1-m}{2} \rightarrow n = m$

f is onto : let $y \in \mathbb{Z}$

If $y > 0$, then $x = \underbrace{2y}_{\text{even}} \in \mathbb{N}$, so

$$f(x) = f(2y) = \frac{2y}{2} = y$$

If $y \leq 0$, then $x = \dots$

$$f(x) = y$$

$$\begin{aligned} y &= f(x) \\ y &= \frac{1-x}{2} \\ x &= 1-2y \end{aligned}$$

Def A set is countable if it $\in \mathbb{N}$ has the same cardinality as $\mathbb{N}$.

Prop If a set A is countable, the elements of A can be written in a line, 1-D array, list, (with a beginning)

$\mathbb{N}$: 1, 2, 3, 4, 5, ...

$\mathbb{Z}$: 0, 1, -1, 2, -2, 3, -3, ...

Proof If a set A is countable, then there is a bijection $f: \mathbb{N} \rightarrow A$, we can let

$$A = \{ f(1), f(2), f(3), \dots \}$$

Corollary An infinite subset of a countable set is countable.

Proof: $A: a_1, a_2, a_3, a_4, a_5, \dots$
 $(B \subseteq A)$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 $b_1, b_2, b_3, \dots$

Prop $\mathbb{Q}$ is countable.

rational #s: fractions of integers

Proof (nonnegative rational #s)

	0	1	2	3	4	...	$\nwarrow \frac{n}{m}$
1	0/1	1/1	2/1	3/1	4/1	...	
2	0/2	1/2	2/2	3/2	4/2	...	

$$\begin{array}{c|cccccc} 3 & \cancel{0/3} & \cancel{1/3} & 2/3 & 3/3 & 4/3 & - & - & - \\ 4 & \cancel{0/4} & \cancel{1/4} & 2/4 & 3/4 & 4/4 & - & - & - \\ \vdots & & & & & & & & \end{array}$$

↓
Q can be written is 1D array/
line

↓
Q is countable

Prop The interval $[0, 1)$ is uncountable

Proof Suppose $[0, 1)$ is countable,
then the elements can be written
in a list :

$$a_1 = .a_{11} a_{12} a_{13} a_{14} \dots$$

$$a_2 = .a_{21} a_{22} a_{23} a_{24} \dots$$

$$a_3 = .a_{31} a_{32} a_{33} a_{34} \dots$$

$$a_4 = .a_{41} a_{42} a_{43} a_{44} \dots$$

⋮

$$\text{Let } b = .b_1 b_2 b_3 b_4 \dots \text{ s.t. } b \in [0, 1) \quad b_k \neq a_{kk}$$

but $b \neq a_i$ for any i $\rightarrow \leftarrow$
Contradiction

Corollary: $\mathbb{R}$ is uncountable

Proof: - - -

HW 9 #2 show $(0, 1]$ and

$[1, \infty)$ have same cardinality



Have to find bijection $f: (0, 1] \rightarrow [1, \infty)$

$$f(x) = \frac{1}{x}$$

f is 1 to 1: $f(x) = f(y)$

$$\rightarrow \frac{1}{x} = \frac{1}{y} \rightarrow x = y$$

f is onto: let $y \in [1, \infty)$

$$\text{then } x = \left(\frac{1}{y}\right) \in (0, 1]$$

$$\text{and } f(x) = f\left(\frac{1}{y}\right) = \frac{1}{1/y} = y$$